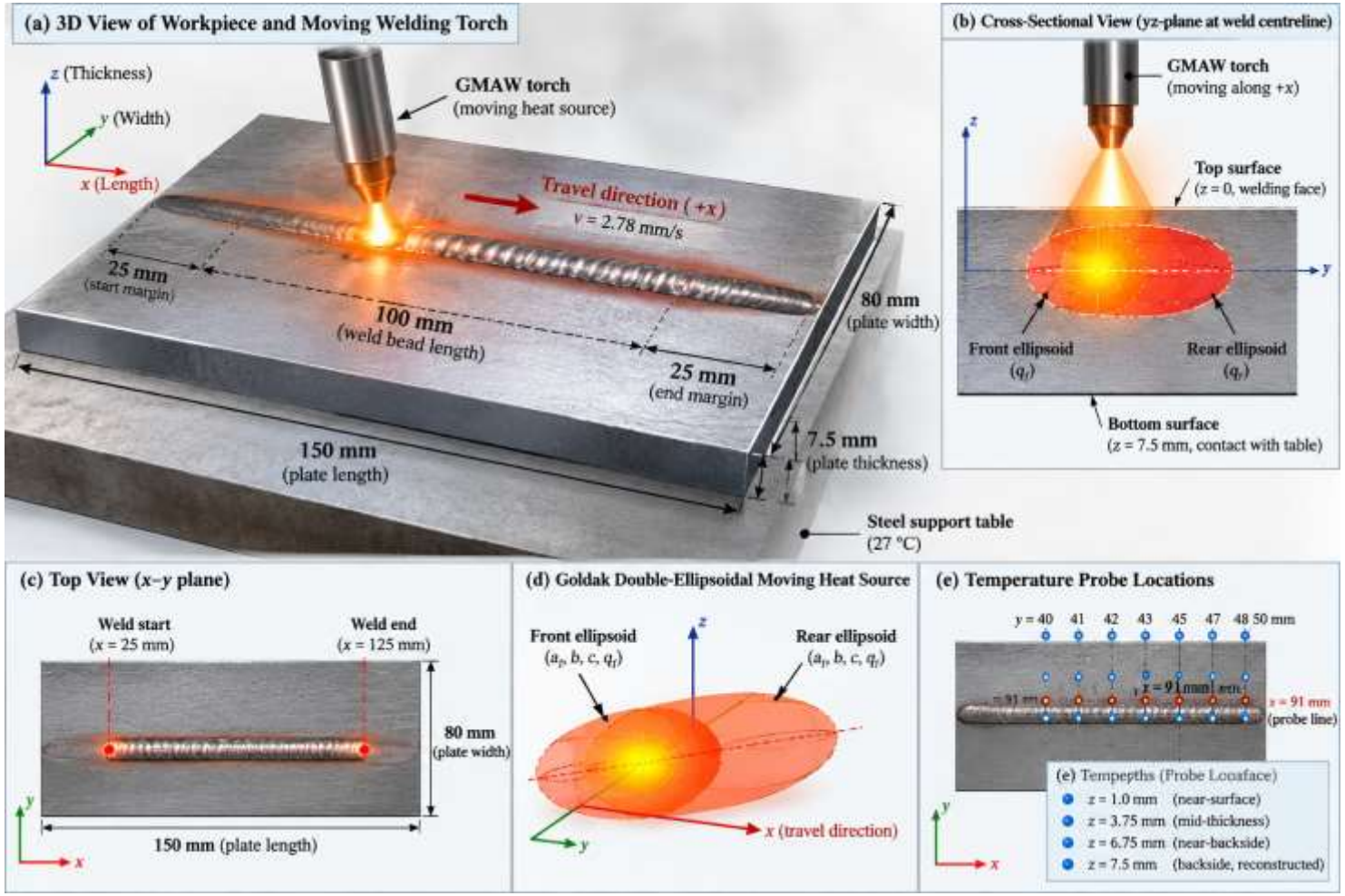
Physics-Based Modelling, Interactive Visualization and Computational Analysis of Moving Heat-Source Welding Processes

(An Interactive Digital Twin Platform for Customized Welding Thermal Simulation)

INTRODUCTION TO WELDING THERMAL SIMULATION



Challenges involved in Welding Thermal Simulation

- ✓ Involved a moving heat source
- ✓ Transient heat transfer and high temperature gradients
- ✓ Complex loading and boundary conditions
- ✓ Welding temperature varies with welding power, travel speed, heat-source characteristics and material properties.

REFERENCE WELDING CONDITIONS, WORKPIECE GEOMETRY AND HEAT-SOURCE PARAMETERS

The initial welding conditions, workpiece geometry and heat-source parameters used in the SimuTwin platform were adopted from the published study by Rocha et al. (2018).

Welding conditions

- Welding process: GMAW
- Current: 192 A
- Voltage: 18.4 V
- Welding speed: 2.78 mm/s
- Thermal efficiency: 85%
- Net heat input power: 2.997 kW

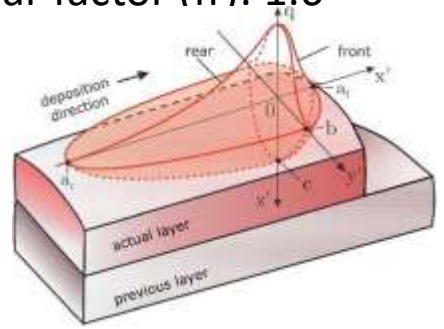
Workpiece geometry

- Joint configuration: Butt joint
- Workpiece dimensions: 150 × 80 × 7.5 mm
- Bevel angle: 45°
- Weld bead length: 100 mm
- Welding direction: Longitudinal



Goldak heat-source parameters

- Transverse semi-axis (b): 4.472 mm
- Depth semi-axis (c): 6.050 mm
- Front semi-axis (af): 3.025 mm
- Rear semi-axis (ar): 12.100 mm
- Front factor (ff): 0.4
- Rear factor (fr): 1.6

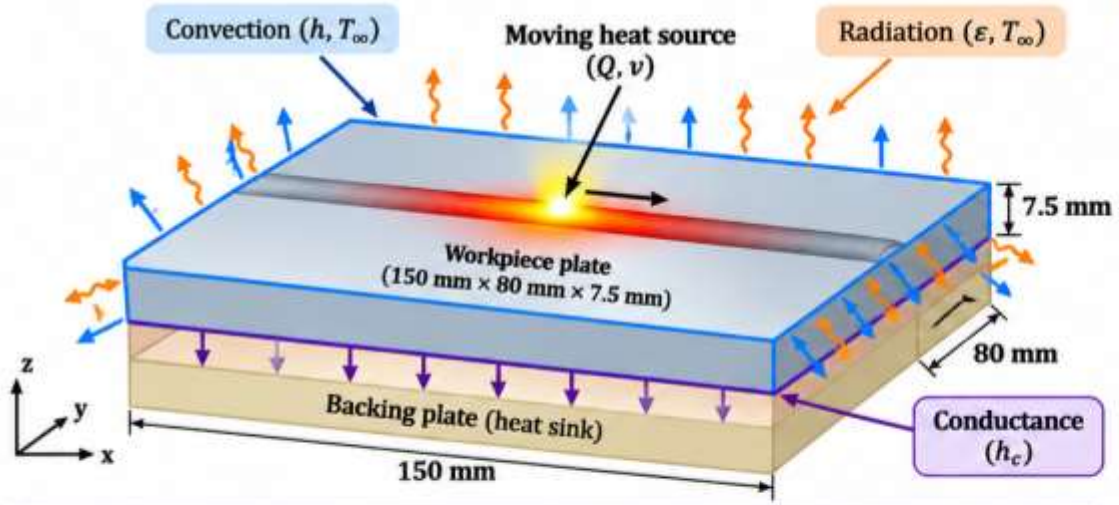


Reference: Rocha, E. J. F., Antonino, T. de S., Guimarães, P. B., Ferreira, R. A. S., Barbosa, J. M. A., & Rohatgi, J. (2018). Modeling of the temperature field generated by the deposition of weld bead on a steel butt joint by FEM techniques and thermographic images. Materials Research, 21(3), e20160796. https://doi.org/10.1590/1980-5373-MR-2016-0796)

LOADING & BOUNDARY CONDITIONS

Loading Condition: Moving Heat Source

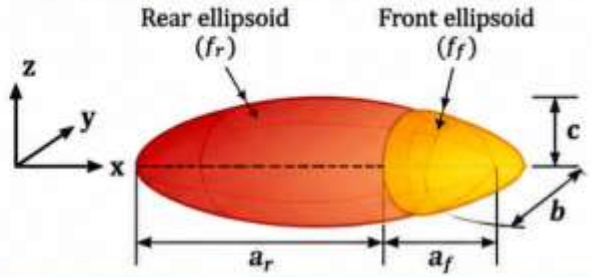
- Goldak double-ellipsoidal heat source
- Net heat input power: $Q = \eta VI = 2.997$ kW
- Welding speed: $v = 2.78$ mm/s
- Heat source moves along the weld line ($x-$direction)



Key Highlights

- **Thermal loading:** Goldak moving heat source, $Q = 2.997$ kW, $v = 2.78$ mm/s.
- **Heat losses:** convection and radiation on all exposed surfaces (top and sides).
- **Bottom boundary:** heat transfer to a backing plate modeled using conductance (h_c).
- **Initial condition:** uniform ambient temperature (T_0).

Goldak Double-Ellipsoidal Heat Source



Boundary Conditions

Convection (all exposed surfaces)

$q_{conv} = h(T - T_\infty)$

- Heat transfer coefficient: h
- Ambient temperature: T_∞

Radiation (all exposed surfaces)

$q_{rad} = \varepsilon\sigma(T^4 - T_{\infty}^4)$

- Emissivity: ε
- Stefan–Boltzmann constant: $\sigma = 5.67 \times 10^{-8}$ W·m^{-2}·K^{-4}

Conductance (bottom interface)

$q_{cond} = h_c(T - T_b)$

- Interface conductance: h_c
- Backing plate temperature: T_b

Initial Condition

$T(x, y, z, t = 0) = T_0$

- Uniform initial temperature: T_0

Reference: Rocha, E. J. F., Antonino, T. de S., Guimarães, P. B., Ferreira, R. A. S., Barbosa, J. M. A., & Rohatgi, J. (2018). Modeling of the temperature field generated by the deposition of weld bead on a steel butt joint by FEM techniques and thermographic images. Materials Research, 21(3), e20160796. https://doi.org/10.1590/1980-5373-MR-2016-0796)

FINITE VOLUME METHOD

A Cell-centred Explicit Finite Volume Method approach is used to do this analysis:

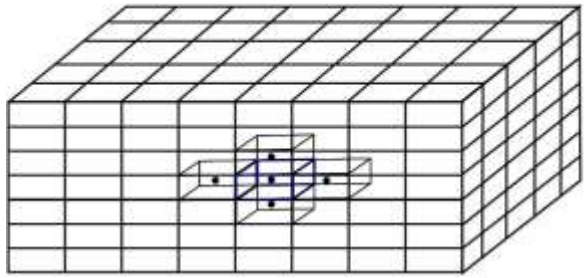
- ✓ Three-dimensional Cartesian computational domain.
- ✓ Cell-centred temperature storage.
- ✓ Energy conservation applied to individual control volumes.
- ✓ Heat conduction calculated through control-volume faces.
- ✓ Explicit time integration for transient temperature evolution.
- ✓ Moving Goldak volumetric heat source incorporated into the energy balance.

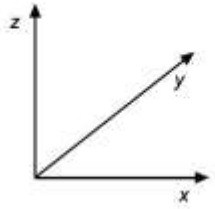
Three-dimensional heat conduction Governing equation

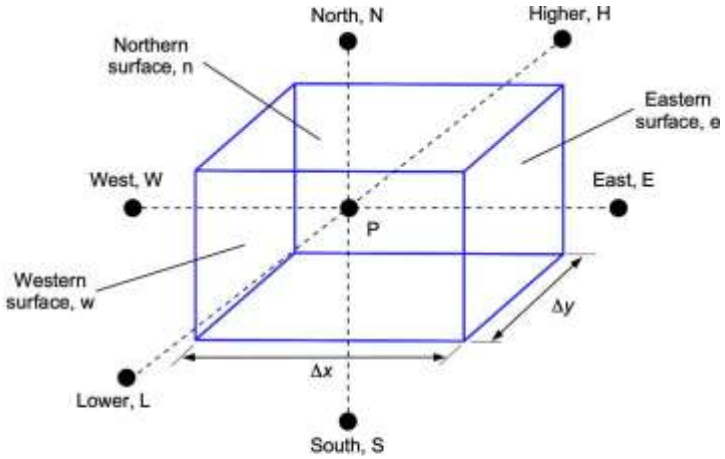
$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \dot{Q}$$

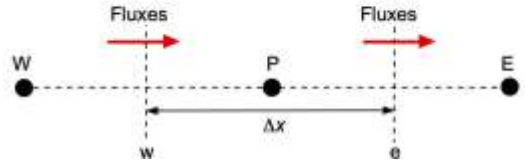
- ρ – Density (kg/m^3)
- C – Specific heat capacity (J/kg K)
- k – Thermal conductivity (W/m K)
- T – Temperature (K)
- $\dot{Q}$ – Volumetric heat generation (W/m^3)

Numerical approach









FINITE VOLUME METHOD

Finite volume discretization

For each computational control volume:

$$\rho C_p V_P \frac{T_P^{n+1} - T_P^n}{\Delta t} = \sum_f k_f A_f \left(\frac{\partial T}{\partial n}\right)_f + \dot{Q}_P V_P$$

Where, V_P is the control-volume size, A_f is the face area and the summation represents conductive heat fluxes across the control-volume faces. Boundary heat losses and heat transfer through the bottom contact interface are incorporated through the corresponding boundary fluxes.

integration and numerical controls

Spatial discretization	Cell-centred FVM
Time integration	Explicit
Material properties	Constant during each simulation
Heat-source model	Goldak double ellipsoid / uniform surface source
Time-step selection	Fixed or adaptive
Stability control	Explicit stability restriction
Mesh	User-defined 3D Cartesian grid
Energy monitoring	Energy balance evaluation

FINITE VOLUME METHOD

3D Computational Domain
Discretization into control volumes

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Finite Volume Energy Balance
Conduction + heat source + boundary fluxes

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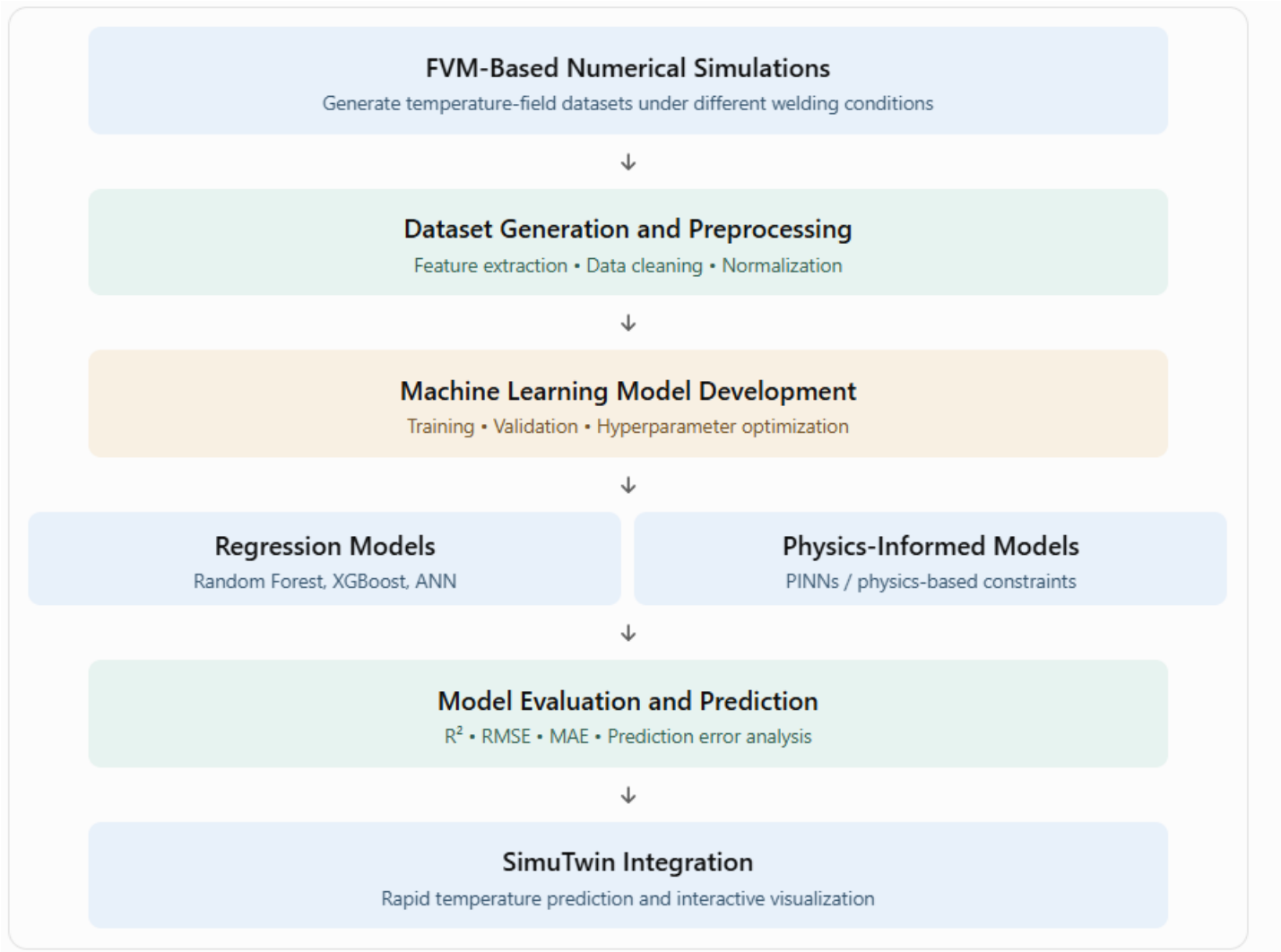
Explicit Time Integration
Temperature update at each time step

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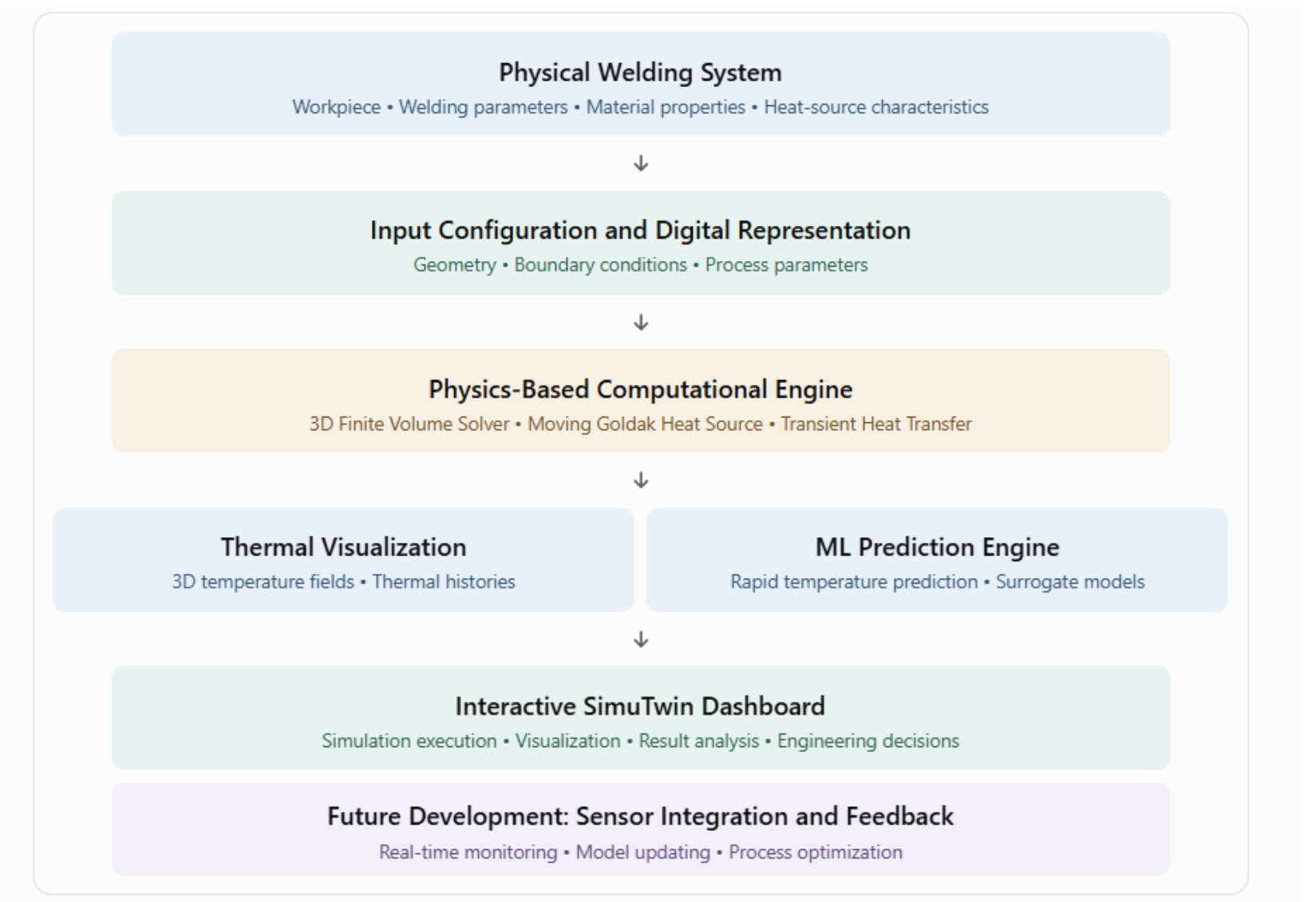
Transient Temperature Field
Temperature distribution at selected times

- FVM calculates heat transfer through the faces of individual control volumes.
- Explicit time integration captures temperature evolution as the welding heat source moves along the workpiece
- Time-step restrictions and mesh-resolution diagnostics are incorporated to improve numerical reliability.

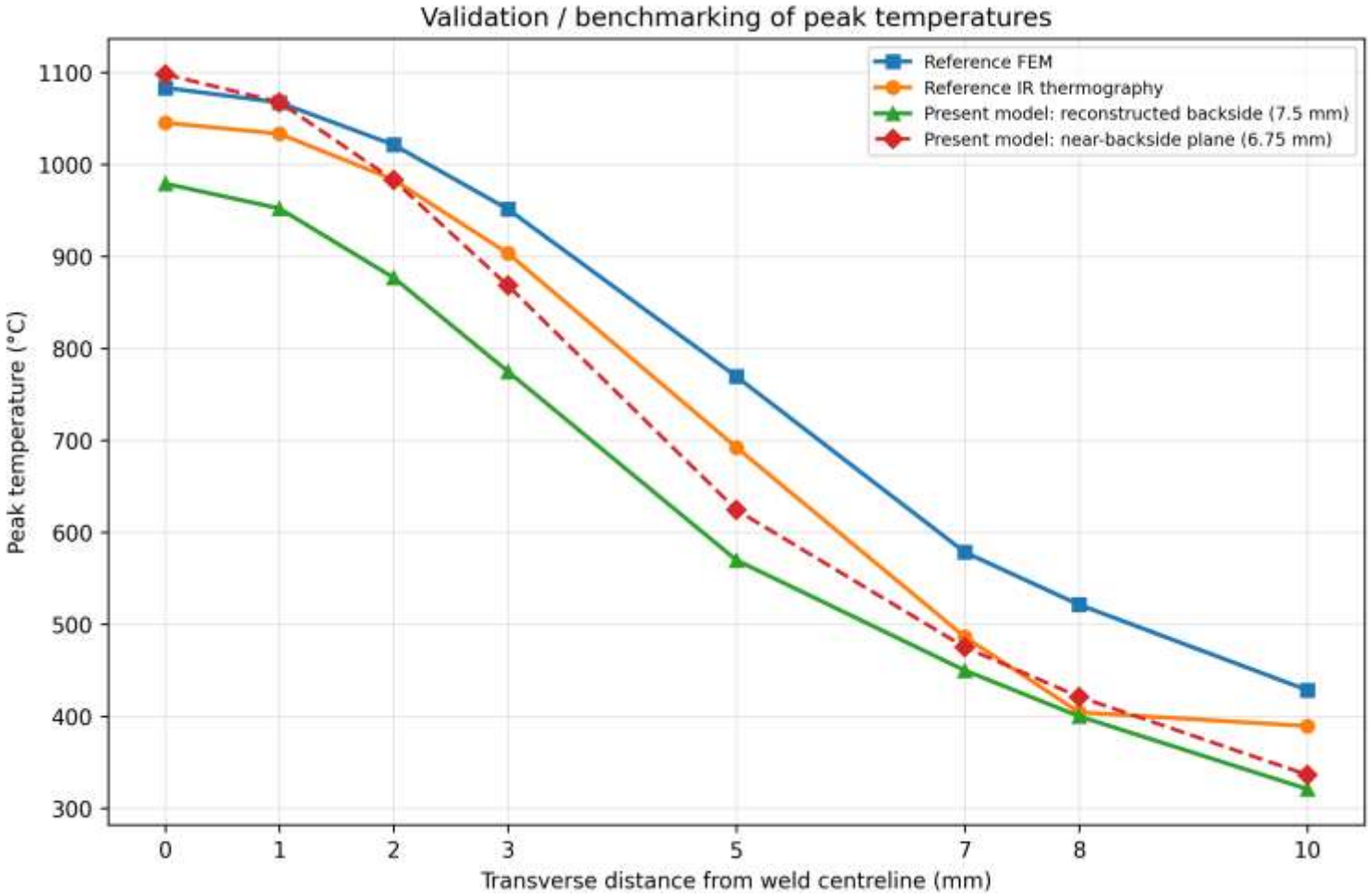
MACHINE LEARNING FRAMEWORK



DIGITAL TWIN FRAMEWORK



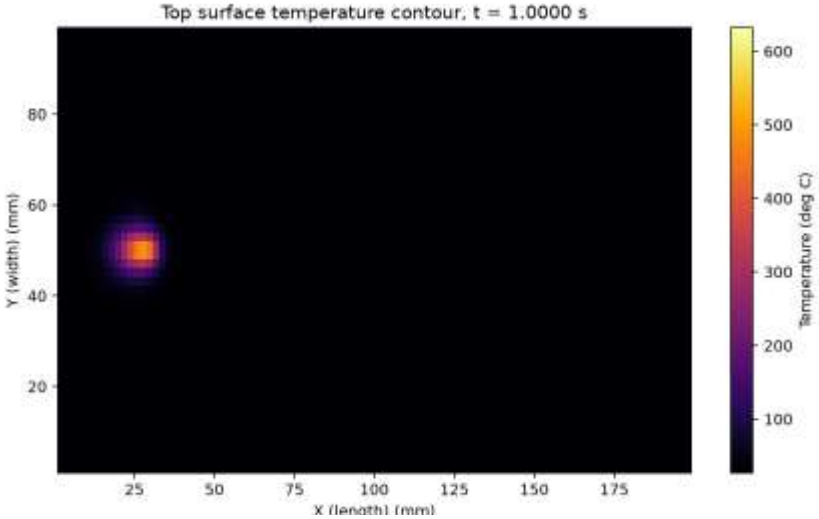
VALIDATION RESULTS

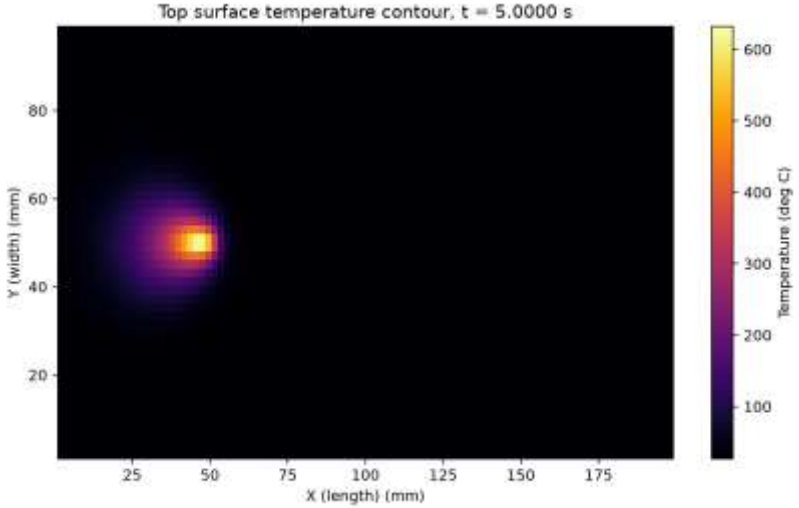


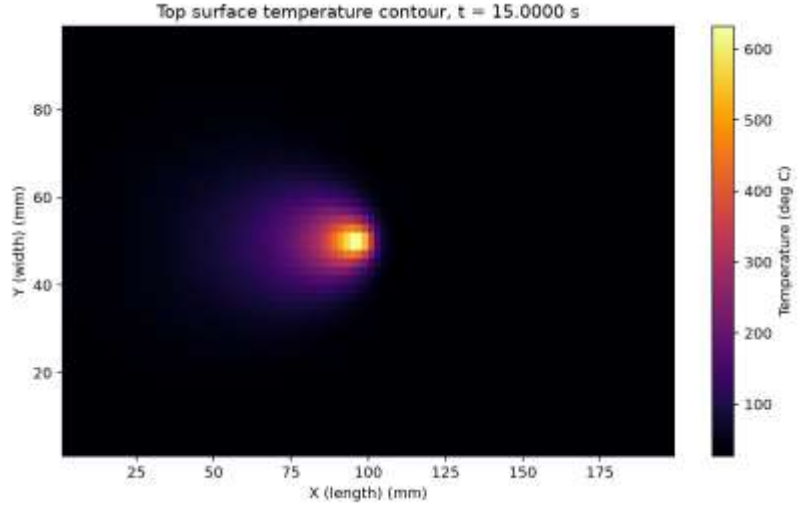
Comparison of peak temperatures predicted by the present Goldak double-ellipsoidal heat-source model (Present FVM method) with reference finite element (FEM) and infrared thermography (IR) results at different transverse distances from the weld centerline, including predictions at 6.75 mm depth and the reconstructed backside surface (7.5 mm).

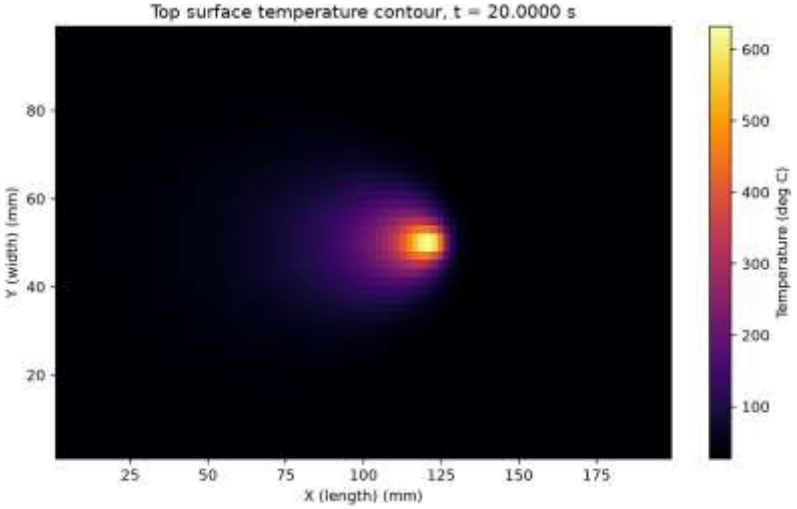
RESULTS & DISCUSSION

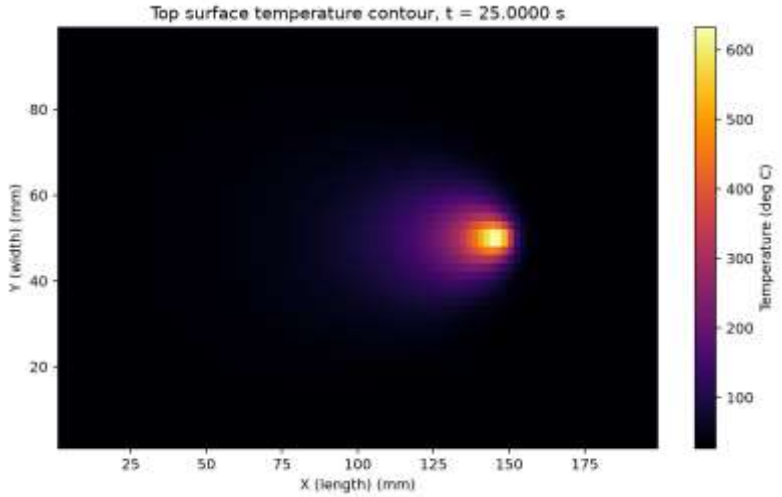
- **WELDING CONDITIONS, WORKPIECE GEOMETRY AND HEAT-SOURCE PARAMETERS, temperature contours during welding at different simulation times (t = 1, 5, 15, 20, 25 and 30 s), illustrating the movement of the localized thermal field along the weld path.**

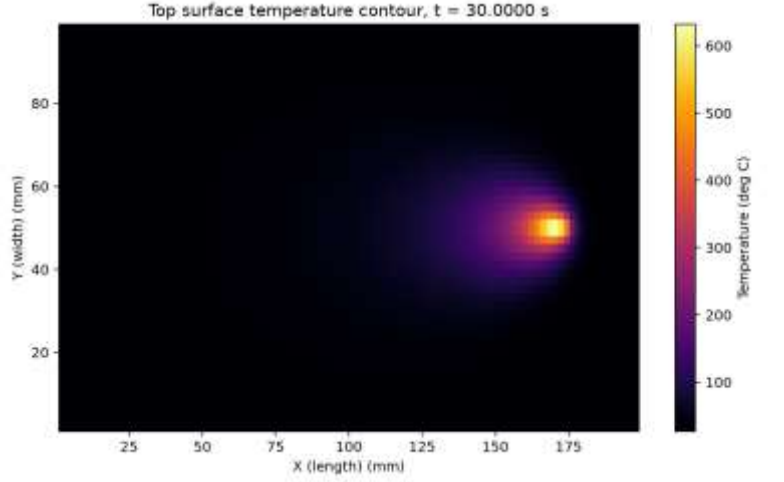






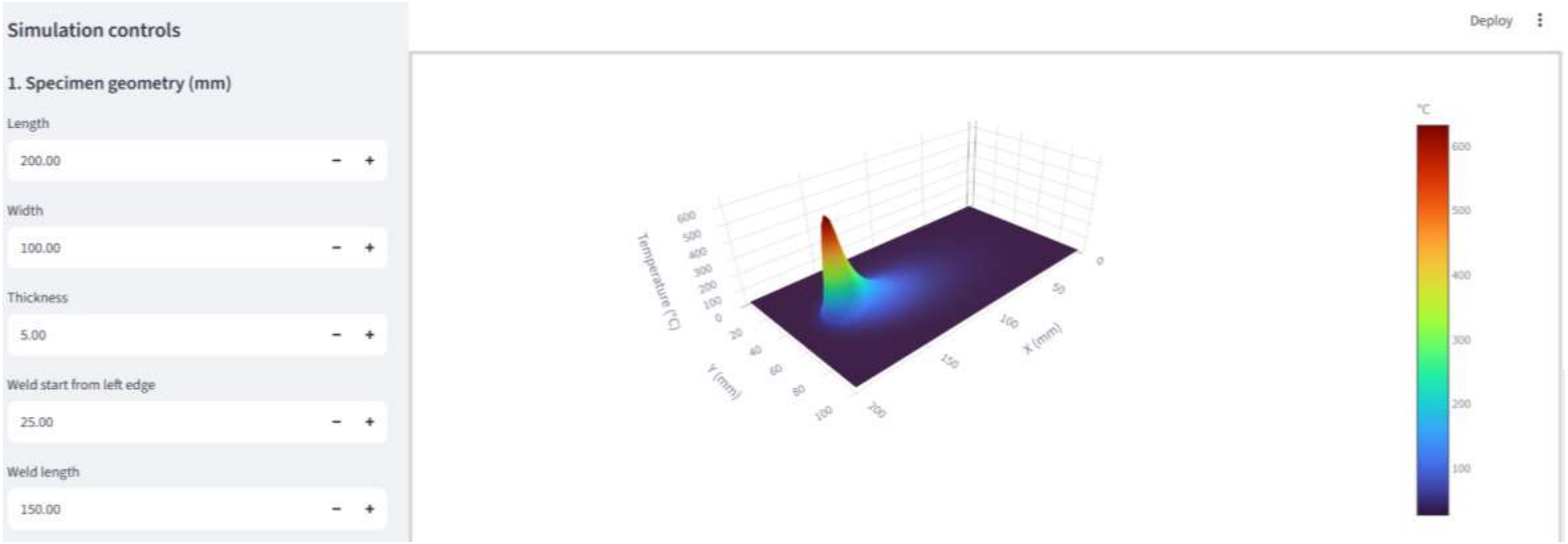






RESULTS & DISCUSSION

Customized digital twin interactive digital twin interface showing user-defined welding parameters, three-dimensional workpiece configuration and the simulated transient temperature distribution.



CONCLUSIONS

1. **Development of a Physics-Based Thermal Solver:** A three-dimensional, cell-centred explicit Finite Volume Method (FVM) solver was developed to simulate transient heat transfer during moving heat-source welding processes using the Goldak double-ellipsoidal heat-source model.
2. **Implementation of Realistic Welding Conditions:** Literature-derived welding parameters, workpiece geometry and heat-source characteristics were incorporated, together with convection, radiation and bottom-interface conductance boundary conditions, to establish a representative welding simulation environment.
3. **Numerical Validation:** The predicted peak temperatures were compared with reference finite element (FEM) and infrared thermography (IR) results at different transverse distances from the weld centreline, including through-thickness and backside temperature predictions. This provides a basis for assessing the predictive capability of the developed solver.
4. **Successful Transient Thermal Visualization:** The simulated temperature contours demonstrate the movement of the localized thermal field along the weld path and its time-dependent spatial evolution, providing insight into the thermal behaviour of the welding process.
5. **Development of an Interactive DIGITAL Twin Platform:** A customized, browser-based interface was developed by integrating the FVM solver with interactive input configuration and three-dimensional temperature visualization, enabling users to configure welding conditions and perform customized thermal simulations.

LIMITATIONS & FUTURE SCOPE

- **Constant Material Properties: The present solver assumes constant thermophysical properties, without accounting for their temperature-dependent variation.**
- **Simplified Thermal Modelling: The current formulation focuses on transient heat transfer and does not fully incorporate metallurgical transformations, phase changes, residual stresses or welding distortion.**
- **Computational Requirements: The explicit FVM formulation requires appropriate mesh resolution and time-step restrictions to maintain numerical stability.**
- **Limited Digital Twin Synchronization: The current platform performs user-configured, batch-based simulations and does not yet support continuous real-time synchronization with physical welding equipment.**


Thank You